NAME

mbm_trnplot − Macro to plot map views of an **mbtrnpp** terrain relative navigation (TRN) session.

VERSION

Version 5.0

SYNOPSIS

mbm_trnplot [**−i***inputdirectory* **−g***gridfile* **−j***crs* **−−mission=***missionname* **−d** **−h**]

DESCRIPTION

mbm_trnplot generates map-view plots comparing an **mbtrnpp** terrain relative navigation (TRN) session's estimated vehicle track against reference bathymetry, using **mbm_grdplot**, **GMT**, and (for a second, locally regridded comparison) **mbgrid**.

The macro searches an input session directory (**−i**/**−−input**; by default the directory pointed to by a symbolic link named "mbtrnpp-latest" in the current directory if one exists, otherwise the current directory) for **mbtrnpp** session log files matching the pattern "*_trn.txt". For each session log found, the macro determines a reference topography grid to plot the track over. This grid can be given explicitly with **−g**/**−−grid**; otherwise the macro looks for a symbolic link named "mbtrnpp-grid" in the current directory, and failing that it parses the session log's own header comments for a line beginning "## Reference topography model:" to locate the TRN octree/grid used during the session (translating a ".bo" octree reference to the corresponding "source_grid.grd" grid, or otherwise appending "/source_grid.grd" to a bare tileset directory name).

Each session log's data lines are parsed to extract, per ping: the navigation-only position, the TRN-estimated position, and lon/lat equivalents of both when present. TRN offset estimates flagged in the log as "CNV", "USE", or "RNT" are additionally written out as separate subsets so that converged, actively-used, and reinitialized fixes can be distinguished on the plot. These extracted positions are written to a set of companion text files in the input directory (with suffixes such as "_trn_Nxy.txt", "_trn_Txy.txt", "_trn_TxyC.txt", "_trn_TxyU.txt", "_trn_TxyR.txt", and lon/lat equivalents ending in "Nll"/"Tll"/etc.), and the overall bounding box of the plotted positions (padded by 10%) is computed.

mbm_trnplot then constructs and runs an **mbm_grdplot** command to generate a shaded-relief map of the reference topography with the navigation-only track, the raw TRN-estimated track, and the CNV/USE/RNT position subsets overlaid, converts the resulting PostScript plot to a JPEG image with **gmt psconvert**, and (unless **−d**/**−−display** is given, in which case the plot is also displayed immediately) runs the generated plotting script in non-interactive mode. The macro then repeats this process a second time against a bathymetry grid built on the fly from any MB1-format (".mb1") processed data files found in the input directory, using **mbm_makedatalist**, **mbdatalist**, and **mbgrid** (with the projection set by **−j**/**−−crs**), so that TRN performance can also be compared against bathymetry gridded directly from the vehicle's own logged soundings.

The plot title uses the mission name given with **−−mission** if supplied, otherwise a session name derived from the log filename.

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OPTIONS

−h, **−−help**

This "help" flag causes the macro to print out a description of its command line options and then exit immediately.

−d, **−−display**

Causes each generated plot to be displayed interactively as soon as it is produced. If this option is not given, the plotting scripts are instead run in non-interactive (batch) mode.

−i*inputdirectory*, **−−input=***inputdirectory*

Sets the path to the directory containing the **mbtrnpp** session log(s) and associated processed results to be plotted. If this option is not given (or is given the literal value "mbtrnpp-latest"), the macro uses the directory pointed to by a symbolic link named "mbtrnpp-latest" in the current directory if one exists, and otherwise uses the current directory.

−g*gridfile*, **−−grid=***gridfile*

Sets the path to the reference topography grid plotted beneath the TRN track. If this option is not given (or is given the literal value "mbtrnpp-grid"), the macro looks for a symbolic link named "mbtrnpp-grid" in the current directory, and failing that attempts to determine the reference grid from the "## Reference topography model:" header comment recorded in the session log.

−j*crs*, **−−crs=***crs*

Sets the coordinate reference system code passed to **mbgrid**'s **−J** option when regridding the session's MB1-format processed data for the second comparison plot. Default: *crs* = "U" (an automatically selected UTM projection).

−−mission=*missionname*

Sets a mission name used in the plot titles in place of the session name derived from the log filename. Note: in the installed version of this macro the short option **−m** does not itself take an attached value (a current limitation of its command line option parsing); use the long form **−−mission=***missionname* **to set this value.**

SEE ALSO

mbsystem(1), **mbtrnpp**(1), **mbm_grdplot**(1), **mbgrdviz**(1), **mbgrid**(1)

BUGS

Maybe. Maybe not.